

Exercise 2

Exercise 2.1 — Safety Vocabulary

1. Loss

A loss is an unacceptable outcome that results in harm to people, property, or the environment.

Example (Autonomous Driving):

A pedestrian is hit because the perception system failed to detect them.

This results in injury or loss of life and is therefore unacceptable.

2. Hazard

A hazard is a system state or condition that can lead to a loss under certain environmental conditions.

Example:

The pedestrian detector fails to recognize a pedestrian crossing the road.

This hazardous condition may eventually lead to collision.

3. Risk

Risk refers to the combination of:

$\text{Risk} = \text{Probability} \times \text{Severity}$

It describes how likely a harmful event is and how severe the consequences are.

Example:

Night driving creates higher risk because pedestrian detection accuracy may decrease.

Exercise 2.2 — ODD Specification

ODD = Operational Design Domain

The ODD defines where and under what conditions the system is intended to operate.

Dimension	Operating Conditions	Non-Operating Conditions	Runtime Detection
Weather	Clear weather, dry roads	Rain, fog, snow	Weather sensor / camera
Lighting	Daytime	Night, low light	Brightness estimation
Camera Condition	Clean lens	Dirt, mud, glare	Image quality check
Scene Type	Urban roads, mapped intersections	Rural roads, highways	Map comparison
Vehicle Speed	≤ 50 km/h	High speed driving	Speed monitoring

ODD Violation Example

If fog appears during operation:

Input distribution changes → perception reliability decreases → system leaves valid ODD.

Exercise 2.3 — Losses

ID	Loss	Description	Why Unacceptable
L-1	Pedestrian injury	Vehicle hits pedestrian	Human safety risk
L-2	Vehicle collision	Collision with another vehicle	Damage and injury
L-3	Property damage	Collision with infrastructure	Economic loss
L-4	Traffic disruption	Wrong decisions causing unsafe traffic	Public safety impact

Exercise 2.4 — Hazards

ID	Hazard	Loss	Likelihood	Severity
H-1	Pedestrian not detected	L-1	Medium	High
H-2	Vehicle ahead missed	L-2	Medium	High

ID	Hazard	Loss	Likelihood	Severity
H-3	Incorrect traffic-light detection	L-3, L-4	Medium	Medium
H-4	Human operator delayed	L-1, L-2	Low	High

Example

H-1:

System state:

Pedestrian exists but detector outputs:

No pedestrian

Worst case:

Planner does not brake.

Result:

Loss L-1.

Exercise 2.5 — Control Structure Diagram

Draw this:

Camera

↓

Perception Models

(Pedestrian / Vehicle / Traffic Light)

↓

Distance Oracle

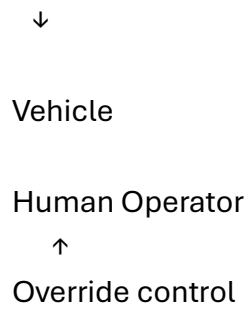
↓

Planning Module

↓

Vehicle Actuators

(Brake / Steering / Throttle)



Add:

Solid arrows = control actions

Dashed arrows = feedback

Save screenshot as:

ControlStructure.png

Exercise 2.6 — Unsafe Control Actions (UCAs)

ID	Controller	Action	UCA Type	Hazard	Unsafe Scenario
UCA-1	Planner	Brake command	Not provided	H-1	Pedestrian detected but braking absent
UCA-2	Planner	Emergency braking	Unsafe provision	H-2	Brakes without vehicle
UCA-3	Human operator	Override	Wrong timing	H-4	Delayed intervention
UCA-4	Planner	Stop command	Wrong duration	H-3	Stops too early

Exercise 2.7 — Safety Constraints

UCA	Safety Constraint	Level	Verification
UCA-1	Pedestrian recall must remain above threshold	Model-level	Testing
UCA-2	Vehicle detection validated before braking	Model-level	Evaluation
UCA-3	Operator alert procedure	System-level	Protocol

UCA	Safety Constraint	Level	Verification
UCA-4	Planner timing verification	System-level	Simulation

Exercise 2.8 — Causal Loss Scenarios

UCA	Scenario	Root Cause	Constraint
UCA-1	Pedestrian missed at night	Distribution shift	SC-1
UCA-2	Vehicle false positive	Label noise	SC-2
UCA-3	Operator fails to react	Fatigue	SC-3
UCA-4	Planner stops too early	Timing delay	SC-4

Exercise 2.9 — Mapping Constraints to Evidence

Constraint Future Evidence

SC-1	Sheet 4 testing metrics
SC-2	Sheet 4 evaluation
SC-3	Safety arguments
SC-4	Simulation validation

FINAL CONCLUSION

The STPA analysis identified hazards, unsafe control actions, and safety constraints for the CARLA autonomous driving system.

Pedestrian detection was identified as the most safety-critical function because missed detections may directly lead to human injury.